

Sahil Padyal

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Skills

Languages: Python, Java Spring Boot, JavaScript, ReactJS, NodeJS, TypeScript, SQL, NextJS
Frameworks & Tools: Kubernetes, Docker, PostgreSQL, GraphQL, Git, SQL, MongoDB, Kafka, Redis, Fast API, OAuth, AWS, GCP
Specialties: Backend, Generative AI, API Development, Cloud Integration, CI/CD, Distributed Systems

Work Experience

Humanitarians AI, United States Jan 2025 - Present
Software Engineer AI/ML

- Contributing to open-source project for Stellis Labs to build custom fine-tuned domain specific LLM models.
- Developing a pipeline to fine-tuning smaller LLMs like **Llama3-1B** and **DeepSeek-1.5B** for domain-specific reasoning.
- Implementing a RAG module to synthesize a database of **5000** entries using open-source models like Llama3-70B via **Groq**.
- Leveraging tools like **Hugging Face Transformers, PEFT, Unsloth, and MLflow** for modular, efficient, and GPU-optimized implementation.
- Building Restful API for the platform using **Python Flask API** and **SQL** database to store metadata and user information.
- Key Skills used:** Transformers, Groq, Python, Unsloth, MLflow, Llama3-1 B, Deepseek-1.5B, HuggingFace

Matrix Rental Solutions, United States May 2023 - Dec 2023
Software Engineer Intern (Generative AI)

- Engineered a high-performance, scalable **AI agent-powered** conversational chatbot using **LangChain** and **Palm-2 LLM** on **GCP Vertex AI**. Integrated advanced reasoning, memory, and tool orchestration to enable dynamic, context-aware interactions, boosting user engagement by **40%**.
- Improved model context accuracy by **60%** through designing and implementing a robust data processing pipeline.
- Increased model accuracy by **46%** by fine-tuning it with **10,000** prompts to handle vague and ambiguous queries effectively.
- Minimized data retrieval time by **20%** through engineering efficient communication between microservices via **REST** and **GraphQL** APIs built on Python **Flask/Fast API**, highlighting expertise in web service architectures.
- Achieved a 40% faster deployment** by enhancing CI/CD workflows, leveraging **Docker** for containerization, and implementing a pipeline with Cloud Build.
- Key Skills used:** GCP, Langchain, Spring Boot, ReactJS, Typescript, Python, Fast API, RESTful APIs, Vertex, SCRUM

Convrse.ai, Singapore Nov 2019 - Jun 2022
Software Engineer Backend

- Spearheaded 0 to 1 development of a 3D mesh optimization tool, leveraging Quadric Error Metrics (QEM) to automatically reduce polygon count while preserving geometric detail.
- Architected a distributed system using **Ray and Redis** for task queuing and caching deployed in **Kubernetes (EKS)** to **handle 100k daily active users**, showcasing proficiency in designing high-performance, scalable backend systems.
- Enhanced system functionality by 25%** through reconstructing the UI using **React.js/Next.js** and **TypeScript**, while implementing a scalable backend with **Python**.
- Optimized database performance and ensured robust data management using **RDS PostgreSQL and MongoDB**, demonstrating expertise in working with relational databases.
- Streamlined deployment processes by developing a CI/CD pipeline using **AWS CodeBuild** and **Argo CD**, reducing deployment time by **50%** and errors by **60%**.
- Cut operational costs by 15%** through strategic architecture transition from **EC2 to Fargate**.
- Ensured data integrity and security with a security layer using **AWS Lambda** for file downloads via CDN
- Key Skills used:** AWS CodeBuild, EC2, Fargate, S3, AWS Lambda, Kubernetes (EKS), ArgoCD, **ReactJS, NextJS, TypeScript**

BlueTickVR, India Jun 2018 - Oct 2019
Software Engineer

- Developed a scalable 3D SaaS inventory tool** for real estate builders using **Java Spring Boot, ReactJS, and WebGL**, generating **\$500,000 in initial revenue** while enhancing 3D visualization and user experience.
- Optimized performance by 40%** through advanced **data structures, algorithms, and OOP**, ensuring high efficiency, early project completion, and robust reliability with **JUnit testing**.

Recent Projects

Warehouse Fleet Dashboard: Developed a Robot Fleet Dashboard leveraging **Kafka** and **Redis**, containerized with **Docker** and **Kubernetes**. [Demo Link](#)

Education

Northeastern University, Boston Sept 2022 – Dec 2024
MS in information Systems